

1D Training Setups: Paper and Augmented

NN_FPN repository notes

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Where the settings live

The 1D training data mode is selected by the YAML field `1d.train.params.training_data`. In the current 1D reproduction config this is `augmented`; in the all-target and smoke configs it is `paper`. The implementation is in `1D/src/train_model.py`: `_training_data_mode`, `_sample_augmented_training_batch`, `_fill_initial_condition`, and `_fill_cross_sections`.

Config / code item	Meaning
<code>training_data: paper</code>	Use the original fixed four-family 1D training set.
<code>training_data: augmented</code>	Draw every training batch from randomized IC/source and material families.
<code>model_tag</code>	Tags checkpoint filenames, e.g. <code>model_augmented_N3_0.pth</code> .
<code>training_time_horizons</code>	Candidate final times for augmented batches. Current value: <code>{0.25, 0.5, 1.0}</code> .
<code>aux_moment_loss_weight</code>	Optional retained-moment penalty added only when <code>obj_idx=0</code> ; current value is 0.

Common numerical/training settings

For the current `configs/reproduce_1d.yaml` run:

Quantity	Value
Domain	$[-1, 1]$, $n_x = 128$, $\Delta x = 1/64$, $\Delta t = \Delta x/2$
Orders	$N \in \{3, 7, 9\}$, exact reference $N_{\text{exact}} = 127$
Optimizer	Adam, learning rate 0.1, NN weight decay 10^{-5} , constant weight decay 10^{-6}
Batch / epochs	batch size 64, epochs 100 in <code>reproduce_1d.yaml</code>
Loss	final-time scalar-flux loss, <code>obj_idx=0</code> ; no auxiliary moment penalty by default

Paper training setup

The paper-mode sampler uses four fixed initial/source families, split across the batch. Cross sections are homogeneous per sample and redrawn each epoch.

Family	Definition on $[-1, 1]$
Gaussian	$\psi_0(x) = \frac{1}{\sqrt{2\pi}0.10} \exp[-x^2/(2 \cdot 0.10^2)]$, no source.
Step	$\psi_0 = 1$ for $ x < 0.2$, no source.
Bump	$\psi_0 = \cos(\pi x)$ for $ x < 0.5$, no source.
Discontinuous source	$\psi_0 = 1$ for $ x < 0.25$; source is 2 for $ x > 0.75$ and 1 for $ x < 0.5$.

For each batch sample in paper mode,

$$\sigma_s \sim U(0, \sigma_{s,\max}), \quad \sigma_a \sim U(0, \sigma_{s,\max} - \sigma_s), \quad \sigma_t = \sigma_s + \sigma_a.$$

With the current config, $\sigma_{s,\max} = 1$. The final time is the fixed training parameter T , not the `training_time_horizons` list.

Augmented training setup

In augmented mode, each batch item independently draws one IC/source family from $\{0, \dots, 6\}$ and one material family from $\{0, \dots, 5\}$. It also draws a final time uniformly from the configured list $\{0.25, 0.5, 1.0\}$.

Augmented IC/source families

Family	Randomized definition
0	Gaussian: center $c \sim U(-0.4, 0.4)$, width $w \sim U(0.03, 0.15)$, amplitude $A \sim U(0.5, 2)$, normalized Gaussian $A/(\sqrt{2\pi}w)$.
1	Step: center $c \sim U(-0.35, 0.35)$, half-width $h \sim U(0.08, 0.35)$, amplitude $A \sim U(0.5, 5)$.
2	Cosine bump: center $c \sim U(-0.35, 0.35)$, width $w \sim U(0.2, 0.55)$, amplitude $A \sim U(0.5, 3)$, profile $A \cos(\pi x - c /(2w))$ for $ x - c < w$.
3	Discontinuous source-like: local block with center $c \sim U(-0.25, 0.25)$, half-width $h \sim U(0.05, 0.25)$, IC amplitude $U(0.5, 2)$; outer source amplitude $U(0.5, 5)$ beyond a random threshold in $[0.65, 0.85]$; local source increment $U(0.25, 2)$.
4	Central block: half-width $h \sim U(0.08, 0.3)$, IC amplitude $U(0.5, 5)$.
5	Two source pulses: each pulse has center $c \sim U(-0.55, 0.55)$, half-width $h \sim U(0.04, 0.16)$, source amplitude increment $U(1, 20)$.
6	Reeds-like source case on $[-1, 1]$: high source $U(5, 50)$ on $x < -0.5$; moderate source $U(0.5, 3)$ on $0.25 < x < 0.55$.

Augmented material / cross-section families

Case	Randomized definition
0	Homogeneous: $\sigma_s \sim U(0, \sigma_{s,\max})$, $\sigma_a \sim U(0, \sigma_{s,\max} - \sigma_s)$, $\sigma_t = \sigma_s + \sigma_a$.
1	Vanishing-like: center $c \sim U(-0.2, 0.2)$, power $p \sim U(2, 6)$, scale $a \sim U(1, 100)$, floor $f \sim U(0, 10^{-3})$, with $\sigma_s = \sigma_t = f + a x - c ^p$.
2	Discontinuous low slabs: high background $H \sim U(0.5, 10)$, low slab value $L \sim U(0, 0.25)$, two slab centers near $\pm U(0.3, 0.7)$, half-width $U(0.05, 0.2)$, and $\sigma_t = \sigma_s + U(0, 1)$.
3	Vacuum window: background $\sigma_s \sim U(0.2, 3)$, $\sigma_t = \sigma_s + U(0, 2)$, then a central window sets both σ_s and σ_t to zero.
4	Strong material inclusion: background total $U(0.05, 1)$, strong total $U(5, 50)$, scatter fraction $U(0, 0.95)$, random material center and half-width.

Are augmented cross sections similar to test-time cross sections?

Yes, mostly by design, but the augmented sampler is broader and not an exact copy of every test case.

Test case	Test-time cross section	Augmented relation
Gaussian	Homogeneous $\sigma_s = \sigma_t = 0.5$.	Covered by material case 0 when absorption is near zero and $\sigma_s \approx 0.5$.
Vanishing cross section	$\sigma_s = \sigma_t = 100x^4$.	Directly similar to case 1; exact test form is inside the range when $c = 0, p = 4, a = 100, f = 0$.
Discontinuous cross section	Background 1, low slabs 0.2, with $\sigma_s = \sigma_t$.	Similar to case 2; locations and widths are randomized. Case 2 often has $\sigma_t > \sigma_s$, so it is broader than the test.
Reeds	Piecewise $(0, 50), (0, 5), (0, 0), (0.9, 1), (0.9, 1)$ on $[0, 8]$, with sources 50 and 1.	Similar to case 5 after mapping the geometry to $[-1, 1]$, with material values scaled by 0.5–1.5 and randomized source strengths.

The main mismatch is that the augmented setup includes additional material regimes that are not used as named test cases: vacuum windows, strong absorbing/scattering inclusions, shifted vanishing profiles, and randomized absorption gaps where $\sigma_t > \sigma_s$. This makes the augmented data a superset-style stress sampler, while the paper mode is much closer to the original homogeneous-training setup.